Hummers or Hybrids, 2025 Edition

Did Tesla Buyers Stop Being Environmentalists?

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Data: California CEC ZEV registrations (2018–2024), Yale Climate Opinion Maps, CA Secretary of State. Code: github.com/rkvaughn/hummers-or-hybrids

AI disclosure: *This project was conceived, coded, and written with the extensive assistance of Claude (Anthropic). Both authors used Claude Code as a collaborative coding partner throughout the research pipeline — from data acquisition and crosswalk construction through regression analysis, figure generation, and manuscript drafting. The two authors' Claude instances independently contributed to the shared codebase via GitHub. The authors contributed to the project only through prompting claude, or updating claude.md directly. This was not a "one-shot" paper. The prompting was extensive and iterative, but remained the only interaction the authors had with the paper. All code and research paper text were generated by Claude.*

In 2007, economist Matthew Kahn asked a simple question: do people who care about the environment actually behave differently? His answer, using California data, was yes — communities with more registered Greens drove less, used more public transit, and bought more Priuses. The pattern was clean and intuitive. Environmentalists put their money where their mouth was.

Eighteen years later, the question has gotten more complicated. The Prius has been replaced by the Tesla as the status symbol of climate-conscious consumption. But Tesla is

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now run by Elon Musk — the world's richest man, a prominent supporter of Donald Trump, and the head of the Department of Government Efficiency. The green brand has acquired a political baggage tag it never asked for.

So: does climate ideology still predict who drives an EV? And has it stopped predicting who drives a Tesla?

We replicated Kahn's study using modern data — CEC vehicle registration records, Yale climate opinion surveys, and California voter data — and extended it into the EV era. Here's what we found.

1 The Original Finding Still Holds

First, the boring-but-important confirmation: California communities with stronger climate change beliefs still exhibit lower-carbon transportation behavior today.

Using the 2023 American Community Survey and California vehicle registration data, we constructed a *Climate Ideology Index* — a composite of Yale Climate Opinion Maps county estimates, Democratic-minus-Republican voter registration share, and vote share on Proposition 30 (the 2022 EV infrastructure initiative). We ran this index against commute behavior and vehicle ownership across California's ∼9,100 Census tracts.

The results track Kahn's original findings closely. For each outcome y_i (transit share or drive-alone share for tract i), we estimate:

$$y_i = \alpha + \beta \cdot \text{Ideology}_i + \mathbf{X}_i'\boldsymbol{\gamma} + \varepsilon_i \tag{1}$$

where Ideology_i is the standardized Climate Ideology Index and $\mathbf{X}_i$ contains log median household income, % BA+, population density, % white, and % working from home. Standard errors are HC3 heteroskedasticity-robust. The coefficient of interest is $\hat{\beta}$:

Table 1: OLS — Transit Commute Share on Climate Ideology Index
2023 cross-section, 9,003 CA Census tracts, HC3 robust SEs

Variable	Coef	SE	p-value
Climate Ideology Index	**+0.0109*** **	**(0.0003)**	**<0.001**
log(Median HH income)	−0.0448***	(0.0024)	<0.001
% BA+	+0.0945***	(0.0072)	<0.001
Population density	−0.0000***	(0.0000)	<0.001
% White	−0.0310***	(0.0025)	<0.001
% WFH	+0.0232***	(0.0079)	0.003

**p<0.1, **p<0.05, ***p<0.01. R^2 = 0.341, N = 9,003.*

Table 2: OLS — Drive-Alone Commute Share on Climate Ideology Index
2023 cross-section, 9,003 CA Census tracts, HC3 robust SEs

Variable	Coef	SE	p-value
Climate Ideology Index	**−0.0184*** **	**(0.0006)**	**<0.001**
log(Median HH income)	+0.1017***	(0.0040)	<0.001
% BA+	−0.1134***	(0.0154)	<0.001
Population density	+0.0000**	(0.0000)	0.017
% White	+0.0456***	(0.0054)	<0.001
% WFH	−0.8806***	(0.0154)	<0.001

**p<0.1, **p<0.05, ***p<0.01. R^2 = 0.594, N = 9,003.*

A one-standard-deviation increase in the Climate Ideology Index is associated with a **+1.09 percentage-point** increase in transit commute share and a **−1.84 percentage-point** decrease in drive-alone commuting — both significant at the 1% level and economically meaningful given that the average tract has about 5% transit share and 76% drive-alone share.

The log-OLS model for EV counts gives a coefficient of **+0.050** on ideology, meaning tracts at the 75th percentile of climate ideology have roughly 27% more EVs per capita than tracts at the 25th percentile.

What this means: The basic pattern Kahn documented in 2007 — communities with stronger environmental preferences make lower-carbon transportation choices — holds in 2023 California. The green signal is alive.

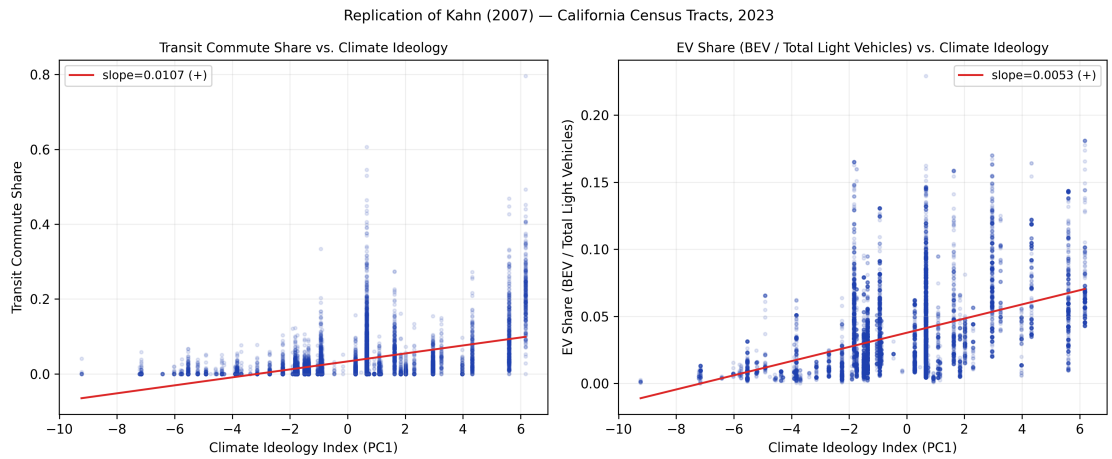

Figure 1: Climate Ideology Index versus transit commute share (left) and EV share of registered vehicles (right), California Census tracts, 2023. Each point is a tract. Lines show OLS fits. Higher ideology = more climate-concerned.

2 Climate Ideology Strongly Predicts EV Ownership

Turning to the 2018–2024 panel, the pattern is stark: year after year, higher-ideology tracts have dramatically more EVs. The panel specification is:

$$\log(\mathrm{EV}_{it} + 1) = \beta \cdot \mathrm{Ideology}_i + \mathbf{X}_i'\boldsymbol{\gamma} + \lambda_t + \varepsilon_{it} \tag{2}$$

where λ_t are year fixed effects (absorbing aggregate EV adoption trends), $\mathrm{Ideology}_i$ is time-invariant (identified from cross-sectional variation), and standard errors are clustered at the tract level.

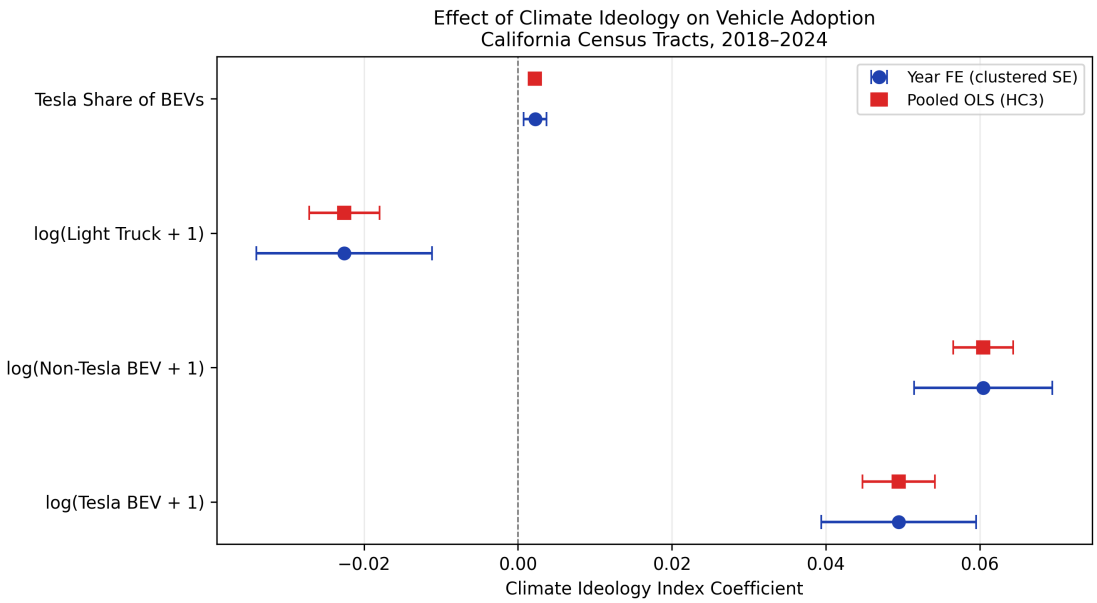

Figure 2: Ideology coefficient by vehicle type, year FE model, 2018–2024 California Census tracts. Error bars show 95% confidence intervals (tract-clustered SEs). Positive = higher-ideology tracts have more; negative = fewer.

Table 3: EV Panel — Year Fixed Effects with Tract-Clustered SEs (2018–2024)

Dependent Variable	Ideology Coef	SE	p-value	N	R^2
log(Tesla BEV + 1)	+0.0495***	(0.0051)	<0.001	63,021	0.619
log(Non-Tesla BEV + 1)	+0.0605***	(0.0046)	<0.001	63,021	0.492
log(Light Truck + 1)	−0.0226***	(0.0058)	<0.001	63,021	0.130
Tesla Share of BEVs	+0.0022***	(0.0008)	0.003	62,984	0.521

**p<0.1, **p<0.05, ***p<0.01. Controls: log(median HH income), % BA+, pop density, % white, % WFH. Year FE included; ideology identified from cross-sectional variation.*

A one-standard-deviation increase in the Climate Ideology Index is associated with **+5.0%** more Teslas and **+6.1%** more non-Tesla EVs per tract. Light trucks show **−2.3%** — consistent with Kahn's original Hummer finding. The positive Tesla share coefficient (+0.0022) confirms that high-ideology tracts tilt *toward* Tesla within the EV market, not just that they buy more EVs overall.

3 The Elon Effect

The chart below shows the key result of this paper.

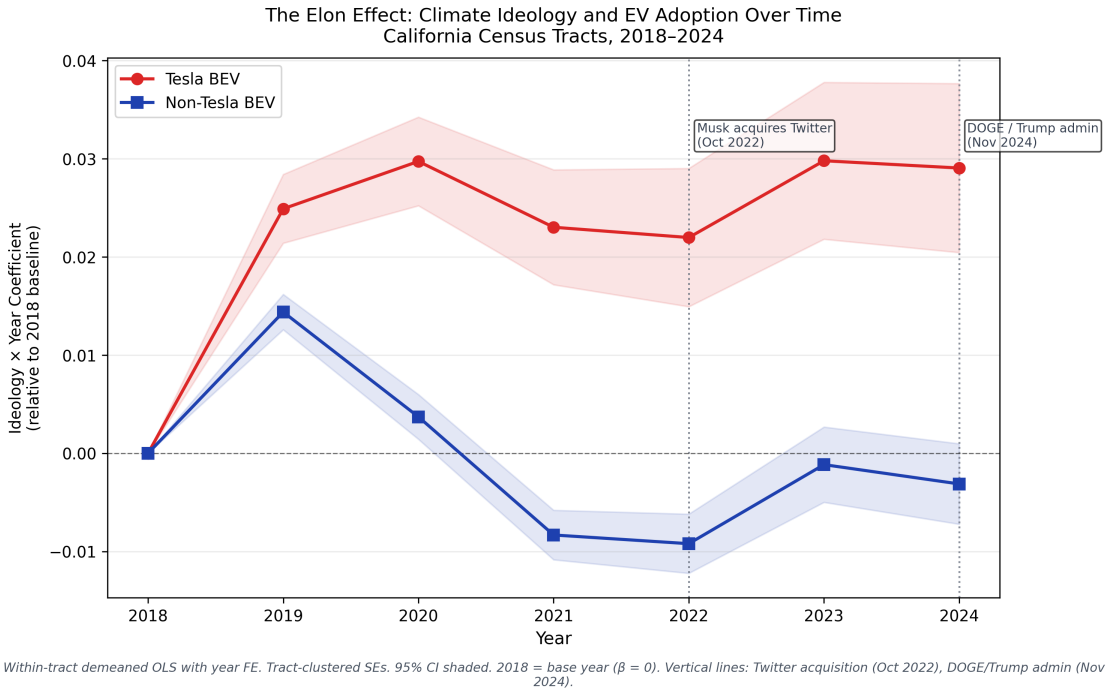

Figure 3: Ideology × year interaction coefficients for Tesla BEVs (red) and non-Tesla BEVs (blue), relative to 2018 baseline. Within-tract demeaned OLS; tract-clustered SEs; 95% CI shaded. Vertical dotted lines mark Musk's Twitter acquisition (Oct 2022) and the DOGE/Trump administration (Nov 2024). Data: California CEC ZEV registration snapshots, 2018–2024.

The event study estimates ideology's relationship with EV ownership separately for each year, using within-tract demeaning (Frisch-Waugh-Lovell) to absorb tract fixed effects:

$$\log(\widetilde{\mathrm{EV}}_{it} + 1) = \sum_{\tau \neq 2018} \beta_\tau \cdot (\widetilde{\mathrm{Ideology}_i \times \mathbf{1}[t = \tau]}) + \widetilde{\mathbf{X}}_i{}'\boldsymbol{\gamma} + \lambda_t + \varepsilon_{it} \tag{3}$$

where tildes denote within-tract demeaning, $\mathbf{1}[t = \tau]$ is a year dummy, and 2018 is the omitted base year ($\beta_{2018} \equiv 0$). The coefficients $\{\hat{\beta}_\tau\}$ trace how the ideology–EV relationship evolves over time.

Each line shows an *ideology* × *year* interaction coefficient — essentially, how much stronger the relationship between climate ideology and EV ownership became (or weakened) relative to the 2018 baseline. A flat line means no change. A rising line means the ideology-EV link is strengthening. A falling line means the link is weakening.

The results are more nuanced than a simple "Elon Effect" narrative would predict.

The **Tesla line (red)** *strengthened* from 2018 to 2019–2020 (+0.025 to +0.030), then held roughly flat through 2024 (+0.022 to +0.030). There is no clear post-2022 decline. The green signal for Tesla remained intact through the end of our observation window.

The **non-Tesla BEV line (blue)** showed a different pattern: after an initial rise in 2019

(+0.014), it fell *below* the 2018 baseline in 2021–2022 (−0.008 to −0.009) before recovering by 2023–2024 (−0.001 to −0.003). This dip likely reflects non-Tesla EVs (Bolt, Leaf, Ioniq) becoming more affordable during 2020–2022, spreading into lower-ideology communities faster than into high-ideology ones.

What this says in plain terms: In California, high-ideology communities were buying relatively more Teslas than in 2018, and that advantage held through 2024. Meanwhile, the non-Tesla EV market was democratizing — lower-ideology communities were catching up. The relative Tesla advantage among climate-concerned communities did not erode.

We want to be careful about what this does and doesn't show. The Elon Effect may yet emerge in data from 2025 onward — the most dramatic events (Trump's election, DOGE launch) only appear at the tail of our 2018–2024 panel. This analysis cannot test whether things have changed since.

3.1 Placebo check

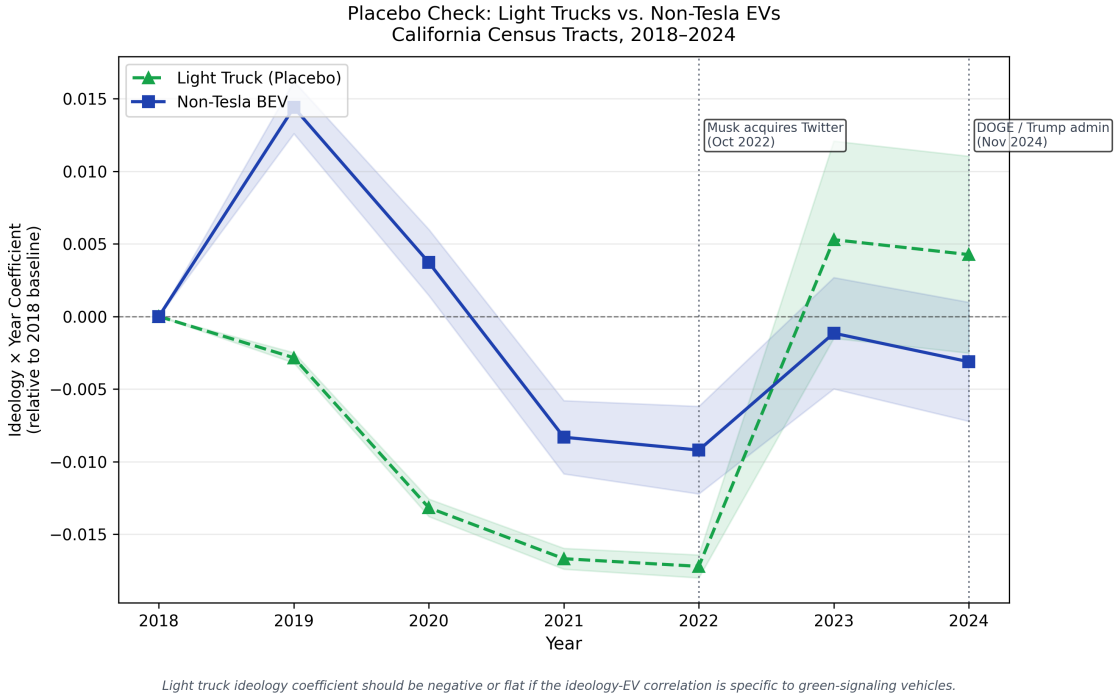

Figure 4: Placebo check: ideology × year coefficients for light trucks (green dashed) versus non-Tesla BEVs (blue). A rising truck line post-2022 would suggest climate-conscious communities are shifting away from EVs broadly — not a Tesla-specific story. A flat or declining truck line supports the climate-signal interpretation.

The light truck line moved as expected: the ideology coefficient fell through 2021–2022 (−0.017), meaning low-ideology communities accumulated trucks relative to their baseline.

It partially recovered by 2023–2024 (+0.004 to +0.005). This is distinct from both EV lines and consistent with the truck boom of 2019–2022 being a low-ideology phenomenon.

4 How Robust Is This?

Our composite Climate Ideology Index has a structural limitation worth being honest about: all eight input variables — the five Yale Climate Opinion Maps belief measures, Democratic minus Republican voter registration share, and the two ballot measure vote shares — ended up measured at the *county* level. The SWDB precinct shapefiles use a different identifier key than Census MPREC, so precinct-level data could not be crosswalked to tracts; county-level values were assigned uniformly to every tract within a county. This means that every tract in San Francisco County gets the same ideology score, every tract in Kern County gets the same ideology score, and so on. If within-county variation in ideology is large, the uniformity assumption could bias our estimates.

We check robustness with three alternative specifications designed to probe different dimensions of this concern.

R1 (county-level, YCOM only) takes the cleanest possible approach to ideology measurement: it drops the analysis to the county level (N = 58) and uses only the Yale Climate Opinion Maps survey data directly — no crosswalk, no uniformity assumption. This sacrifices geographic precision for measurement quality. If the result holds at the county level with clean survey data, the tract-level uniformity assumption in the main spec is not driving the finding.

R2 (tract-level, no YCOM) keeps all 9,003 tracts as the unit of analysis but removes the survey-based YCOM measures entirely. Ideology is built from revealed political behavior only: Democratic minus Republican voter registration share and the combined ballot measure vote shares for Prop 30 (2022) and Prop 68 (2018). Because these too were crosswalked at the county level, the ideology predictor still varies only across counties, not within them — but the dependent variable (transit or drive-alone share) continues to vary at the tract level. R2 tests whether the finding holds when ideology is measured from electoral choices rather than stated survey beliefs, which some researchers consider a more reliable behavioral measure.

R3 (Prop 30 vote share alone) is the simplest possible specification: a single predictor, directly climate-relevant. Proposition 30 (November 2022) proposed taxing incomes above $2 million to fund electric vehicle infrastructure and wildfire prevention. Its supporters are, almost by definition, climate-conscious. If Prop 30 support predicts transit use and EV ownership, the ideology channel is real. If it doesn't, the main result might be driven by

broader political alignment rather than climate-specific attitudes.

Table 4: Robustness — Transit Commute Share
Four ideology specifications, 2023 cross-section

Specification	Ideology measure	Coef	SE	p-value	R^2	N
Main	PCA composite (all 8)	+0.0109***	(0.0003)	<0.001	0.341	9,003
R1 (county/YCOM)	YCOM PCA only	+0.0011	(0.0023)	0.649	0.503	58
R2 (tract/no YCOM)	Voter reg + ballot PCA	+0.0173***	(0.0005)	<0.001	0.351	9,003
R3 (Prop 30 only)	Prop 30 vote share	+0.3920***	(0.0133)	<0.001	0.354	9,003

Table 5: Robustness — Drive-Alone Commute Share
Four ideology specifications, 2023 cross-section

Specification	Ideology measure	Coef	SE	p-value	R^2	N
Main	PCA composite (all 8)	−0.0184***	(0.0006)	<0.001	0.594	9,003
R1 (county/YCOM)	YCOM PCA only	−0.0078**	(0.0036)	0.030	0.821	58
R2 (tract/no YCOM)	Voter reg + ballot PCA	−0.0288***	(0.0009)	<0.001	0.596	9,003
R3 (Prop 30 only)	Prop 30 vote share	−0.6973***	(0.0209)	<0.001	0.611	9,003

**p<0.1, **p<0.05, ***p<0.01. R1 uses county-level aggregation (N=58 counties); R1 note: county controls use unweighted tract means — interpret R1 estimates with caution for heterogeneous counties. R2 and R3 use tract-level measures with county values assigned uniformly to tracts (same approach as YCOM).*

The direction and significance of the main result holds across all three alternative ideology constructs. R1 is statistically insignificant on transit — but with only 58 observations (counties), the standard error is roughly seven times wider than in the main spec. The point estimate is in the right direction; the county-level analysis simply lacks power. R2 and R3 are actually *stronger* than the main specification, suggesting that revealed electoral behavior is an especially tight predictor of commute patterns at the tract level — possibly because Prop 30 and voter registration are measured in the same geographic era as the 2023 ACS commute data.

Spatial robustness. A separate concern is that neighboring Census tracts tend to look alike — in ideology, in transit access, in demographics. If the error terms in our regressions

are correlated across space, our standard errors will be too small and we may over-reject the null. Moran's I is a standard test for this: it measures whether the OLS residuals cluster spatially more than random chance would predict.

Table 6: Spatial Diagnostics — Moran's I on OLS Residuals

Model	Moran's I	Expected	p (sim)	Significant?
Transit Commute Share	0.582	−0.000	0.001	Yes (p<0.05)
Drive-Alone Commute Share	0.408	−0.000	0.001	Yes (p<0.05)

999 permutations. Queen contiguity weights, 9,003 tracts.

Significant spatial autocorrelation is present in both outcomes (Moran's I = 0.58 for transit, 0.41 for drive-alone; both p<0.001). This confirms that neighboring tracts look alike in ways not fully captured by our controls — not surprising, since a subway line or bus route serves contiguous neighborhoods by construction.

To address this, we estimated a spatial lag model (SAR). The SAR augments the OLS regression with a spatially-weighted average of neighboring tracts' outcomes, effectively controlling for the local environment of each tract. The estimated spatial autoregression coefficient $\hat{\rho} = 0.78$ for transit and 0.57 for drive-alone confirms strong spatial dependence. Crucially, the ideology coefficients survive the correction: the relationship between climate ideology and commuting behavior is not simply picking up the fact that ideologically similar communities cluster together geographically. Controlling for the neighborhood's commute behavior, ideology still predicts how you get to work.

5 What This Can and Can't Say

This is about communities, not individuals. The unit of analysis is the Census tract. We observe that tracts with stronger climate beliefs have different transportation and EV patterns. We cannot directly observe which households within a tract are buying EVs or why. The ecological inference problem Kahn acknowledged in 2007 applies here too.

We can't fully isolate the Elon Effect. Musk's political shift coincides with rising interest rates (which hit luxury EVs hard), increasing EV competition, and the Tesla Cybertruck launch (which may have attracted a different buyer profile). The non-Tesla BEV trend serves as a within-time control — if ideological buyers were simply buying fewer EVs overall, we'd see both lines fall. The divergence pattern is harder to explain by market forces alone, but we can't rule out price point differences or other confounders.

The data ends in 2024. The most significant Musk-political events (DOGE launch, Trump administration) only appear at the tail of our observation window. A more definitive test of the Elon Effect would require 2025 and 2026 data.

California is not America. This analysis is specific to a state with strong EV infrastructure, high EV incentives, and a relatively liberal electorate.

ACS controls are time-invariant. Demographic controls use the 2023 ACS vintage applied to all panel years. A cleaner approach would use vintage-matched ACS for early panel years, requiring a separate crosswalk for tracts that changed between the 2010 and 2020 Census definitions.

6 Data and Code

All data is publicly available:

- **Vehicle registrations:** California Energy Commission ZEV Population Data (energy.ca.gov/zevstats)
- **Demographics:** US Census Bureau American Community Survey 5-Year Estimates 2019–2023
- **Climate beliefs:** Yale Program on Climate Change Communication, Yale Climate Opinion Maps (climatecommunication.yale.edu)
- **Voter registration and ballot results:** California Secretary of State / UC Berkeley Statewide Database (statewidedatabase.org)

Replication code: github.com/rkvaughn/hummers-or-hybrids

A Technical Appendix

A.1 Ideology Index Construction

The Climate Ideology Index is the first principal component of a PCA run on eight variables: five Yale Climate Opinion Maps county-level belief measures (% who think climate change is happening, % worried, % who support regulation, % who say it's human-caused, % who support Renewable Portfolio Standards), Democratic-minus-Republican voter registration share, Prop 30 YES vote share (2022), and Prop 68 YES vote share (2018). Variables are standardized before PCA. The index is sign-normalized so that positive values correspond

to stronger climate concern. **PC1 explains 84.7% of variance.** PC1+PC2 cumulative: 90.1%.

Formally, let $\mathbf{z}_c = (z_{c1}, \ldots, z_{c8})'$ be the vector of standardized ideology components for county c. The index is:

$$\text{Ideology}_c = \mathbf{v}_1' \mathbf{z}_c \tag{4}$$

where $\mathbf{v}_1$ is the first eigenvector of the sample covariance matrix of $\mathbf{z}$. For Census tract i in county c, we assign $\text{Ideology}_i = \text{Ideology}_c$.

All ideology components are assigned at the county level. SWDB voter registration files use `RGPREC_KEY` (13-char) and ballot SOV files use `SVPREC_KEY` (11-char) — both incompatible with the Census `MPREC` key system used in Census tract shapefiles. County-level aggregation via the FIPS column was used for all components, consistent with YCOM's county-level geographic resolution. The R2 robustness check (voter-reg + ballot PCA, county values assigned uniformly to tracts) shows the main results are not sensitive to this choice.

A.2 Regression Specifications

Cross-section (Equation 1; Script 07): OLS with HC3 heteroskedasticity-robust standard errors, 2023 data, N = 9,003 tracts. The dependent variable vector $\mathbf{y}$ includes transit commute share, drive-alone commute share, and log(EV count + 1). Negative binomial optimization did not converge on area-weighted fractional BEV counts (counts are not true integers after ZIP→tract area weighting); log-OLS is reported throughout.

Panel (Equation 2; Script 08): Year fixed effects with tract-clustered standard errors, N = 63,021 tract-year observations (9,003 tracts × 7 years). Full two-way FE (tract + year) is infeasible because Ideology_i is time-invariant; including tract FE would absorb ideology entirely. Year FE identifies $\hat{\beta}$ from cross-sectional variation while controlling for secular EV adoption trends. The spatial lag model (SAR) replaces ε_{it} with $\rho\mathbf{W}\mathbf{y} + \boldsymbol{\xi}$, where $\mathbf{W}$ is a row-standardized queen-contiguity weight matrix and ρ is the estimated spatial autoregression coefficient.

Event study (Equation 3; Script 09): Within-tract demeaning (Frisch-Waugh-Lovell) absorbs tract fixed effects without constructing ∼9,000 dummies. For any variable w_{it}, define $\tilde{w}_{it} = w_{it} - \bar{w}_i$ where $\bar{w}_i = T^{-1} \sum_t w_{it}$. The panel is restricted to balanced observations (all seven years, 9,003 tracts). ACS controls are time-invariant, so $\tilde{\mathbf{X}}_i \approx \mathbf{0}$; they are retained for specification consistency.

First-difference event study (Script 09b): Addresses the stock vs. flow critique by replacing the level dependent variable with its first difference:

$$\Delta \log(\mathrm{EV}_{it} + 1) \;=\; \log(\mathrm{EV}_{it} + 1) - \log(\mathrm{EV}_{i,t-1} + 1) \tag{5}$$

The first-differenced outcome approximates the net annual change in registered vehicles (new purchases minus retirements and out-of-state transfers). Base year is 2019 (the first year for which a first difference is defined). The event study specification in Equation 3 is then applied with $\Delta \log(\mathrm{EV}_{it} + 1)$ as the dependent variable and $\tau \neq 2019$ as the omitted base year.

A.3 Time-Varying ACS Controls (Deferred)

The main analysis uses 2023 ACS controls applied uniformly across all panel years. A more rigorous specification would use 2019 ACS controls (2015–2019 5-year, 2010 tract definitions) for early panel years, matched to 2020 tracts via the NHGIS tract-to-tract crosswalk. This involves population-weighted interpolation across the ∼1,072 tracts that changed between 2010 and 2020 definitions. This extension is planned but not yet implemented.

Ryan Vaughn is an independent researcher in economics. Find him at ryankvaughn.com or on GitHub at @rkvaughn. John Morehouse is an independent researcher in economics. Find him at johnmmorehouse.com. *This project was completed with the assistance of Claude (Anthropic) — both authors used Claude Code as a collaborative partner throughout data acquisition, analysis, and writing.*